



Coordination View Version 2.0 for IFC 2x3

including it's standard Exchange Requirements

History

Document Code	Title	Organisation/Author	Date	Status
	Coordination View 2.0 with three Exchange Requirements Version 1.6 Final (internal V19)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	26.02.2013	Final
	Coordination View 2.0 with three Exchange Requirements Version 1.5 Final (internal V18)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	12.12.2012	Final
	Coordination View 2.0 with three Exchange Requirements Version 1.4 Final (internal V17)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	05.05.2011	Final
	Coordination View 2.0 with three Exchange Requirements Version 1.3.1 Final (internal V16 new)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	11.03.2010	Final
	Coordination View 2.0 with three Exchange Requirements Version 1.3 Final (internal V16)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	02.03.2010	Final
	Improved Coordination View Version 1.2 Final	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	12.01.2010	Final
	One Coordination View with three Exchange Requirements (internal V15)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	10.02.2010	not published
	Improved Coordination View Version 1.1 Final	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	28.01.2010	Final
	Remove hatching and line types (internal V14)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	26.01.2010	not published
	Remove deprecated entities (internal V13)	Karl-Heinz Häfele Andreas Geiger Thomas Liebich	21.01.2010	not published

	Changes in order to get a valide sub schema for CV Import (internal V12)	Karl-Heinz Häfele Andreas Geiger	10.12.2009	not published
	Improved Coordination View Version 1.0 Final	Karl-Heinz Häfele Thomas Liebich	12.11.2009	Final

Change log

Date	Version	Person	Details	Summary
26.02.2013	Version 1.6	AG	add IfcRelCoversBldgElements for assigning coverings to elements	
12.12.2012	Version 1.5	AG	remove IfcSurfaceCurveSweptAreaSolid, and add IfcSizeSelect for use in IfcCurveStyle	
05.05.2011	Version 1.4	AG	made IfcTypeProduct and IfcBooleanResult non-abstract (allow CSG), remove IfcVibrationIsolatorType and IfcElectricalCircuit	
11.03.2010	Version 1.3.1	AG	added IfcRelAssociates to the list of entities not allowed to be instantiated	no view definition schema change
08.03.2010	Version 1.3 Final	KHH	Post ISG Meeting 17.-18.02.2010 - IfcGroup can be instantiated by all ER --> No schema changes in CV 2.0 - IfcRelConnectsWithRealizingElements added for ER Structural --> No schema changes CV 2.0 - Subschemas of CV V2.0, Architecture, BuildingServices, Structural checked	no view definition schema change
18.02.2010	internal	KHH	ISG Meeting 17.-18-02.2010 Munich: - Naming "Coordination View Version 2.0" for the model view agreed - Change from 4 Coordination Views to 1 Coordination View with 3 Exchange Requirements agreed - Naming "Architecture", BuildingServices" and "Structural" for the exchange requirements agreed	no view definition schema change
12.02.2010	Version 1.2 Final	KHH	Publication as input to ISG Meeting 17.-18.02.2010	
10.02.2010	internal	KHH	Table changed: - No longer four Coordination Views rather than one Coordination View and three Exchange requirements. - Reason: CV-Arch, CV-MEP and CV-Struct are real subsets of CV-Import	
09.02.2010	internal	TL	Adding a wall paper representation of the Coordination View Version 2.0 to the documentation set	development of first valid EXPRESS-G wall paper for a Model View Definition
28.01.2010	Version 1.1 Final	KHH	Publishing updated entity list	
26.01.2020	internal	KHH	Removing of hatching and line types (request from Lassi Lifländer, supported by Thomas Liebich)	schema change
10.12.2020	internal	KHH/AG	First valid schema for CV Import (only for checking purposes).	

AG = Andreas Geiger
KHH = Karl-Heinz Häfele
TL = Thomas Liebich

Scope of this Document:

This document is **not** a **complete** model view definition. It contains only the **ENTITY LIST** of the **Coordination View Version 2.0** and **it's standard Exchange Requirements for IFC 2x3**. Details of the entities are described in additional documents.

Coordination View Version 2.0:

The Coordination View supports [Three Standard Exchange Requirements](#):

- Architecture
- BuildingServices
- Structural

Naming:

Name: Coordination View Version 2.0

Keyword in file Header: CoordinationView_V2.0

Abbreviation: CV V2.0

Abbreviations:

MVD: Model View Definition

ER: Exchange Requirement

MEP: Mechanical, Electrical, Plumbing

ABS: Abstract

Schema:

The Coordination View Version 2.0 is a subset of IFC 2x3 TC1. Therefore the corresponding schema **IFC2x3TC1 must be used**. In order to check the model view definition and the exchange requirements corresponding sub schemata are created and check. These schemata are **only for checking purposes**.

In the Coordination View Version 2.0:

- Actor (partially)
- External Reference (partially)
- Geometric Constraint (partially)
- Geometric Model (partially)
- Geometry (partially)
- **Material (completely)**
- Measure (partially)
- Presentation Appearance (partially)
- Presentation Organization (partially)
- Presentation (partially)
- Profile (partially)
- Property (partially)
- Representation (partially)
- Topology (partially)
- Utility (partially)
- Kernel (partially)
- Product (partially)
- Shared Bld (partially)
- Shared Bldg Service (partially)
- Shared Components (partially)
- Building Controls (partially)
- Electrical (partially)
- Hvac (partially)
- Plumbing Fire Protection (partially)

Out of the Coordination View Version 2.0:

- **Approval (completely)**
- **Constraint (completely)**
- **Cost (completely)**
- **Date Time (completely)**
- **Material Property (completely)**
- **Presentation Definition (completely)**
- **Presentation Dimensioning (completely)**
- **Profile Property (completely)**
- **Quantities (completely)**
- **Structural Load (completely)**
- **Time Series (completely)**
- **Control (completely)**
- **Process (completely)**
- **Construction Management (completely)**
- **Shared Facilities (completely)**
- **Shared Mgm (completely)**
- **Architecture (completely)**
- **Facilities Management (completely)**

Entities, not allowed to be instantiated:

- IfcTypeObject as **Supertype** of IfcTypeProduct
- IfcProductRepresentation as **Supertype** of IfcProductDefinitionShape and IfcMaterialDefinitionRepresentation
- IfcRepresentation as **Supertype** of IfcStyleModel and IfcShapeModel
- **IfcRepresentationContext** as **Supertype** of IfcGeometricRepresentationContext
- **IfcRelAssociates** as **Supertype** of IfcRelAssociatesClassification and IfcRelAssociatesMaterial

Entities, for a valid schema:

- IfcAxis1Placement as **Axis** of IfcRevolvedAreaSolid

Hatching and line style:

- **No Hatching and line style**
- only IfcCurveStyle with CurveColour and CurveWidth (CurveFont is **not** allowed)

Deprecated Entities:

- IfcConnectionPortGeometry
- Ifc2DCompositeCurve
- IfcCraneRailAShapeProfileDef
- IfcCraneRailFShapeProfileDef
- IfcProductRepresentation
- IfcRepresentation
- IfcRepresentationContext
- IfcRelOverridesProperties
- IfcElectricalElement
- IfcElectricalCircuit
- IfcElectricDistributionPoint
- IfcEquipmentElement
- IfcRelCoversSpaces
- IfcGasTerminalType
- IfcVibrationIsolatorType

KIT / IAI / KHH 23.02.2010

Legend:

GREY y(es) : Abstract Entity Entity in View

GREY n(o) : Abstract Entity Entity out of View

GREEN (1): **Entity** in View respectively in Exchange

RED (0): **Entity** out of View respectively out of

View Definition

Coordination View Version 2.0	
Non Abstract	Abstract
556	97
266	61
290	36
556	97

Statistic
All IFC Entities
In
Out
Total In and Out

Exchange Requirements

Architecture		BuildingServices		Structural	
Non Abstract	Abstract	Non Abstract	Abstract	Non Abstract	Abstract
556	97	556	97	556	97
188	49	212	56	172	49
368	48	344	41	384	48
556	97	556	97	556	97

			Coordination View Version 2.0
	Actor	Actor	
	IfcActorRole	Actor	1
	(ABS) IfcAddress	Actor	y
	IfcOrganization	Actor	1
	IfcOrganizationRelationship	Actor	0
	IfcPerson	Actor	1
	IfcPersonAndOrganization	Actor	1
	IfcPostalAddress	Actor	1
	IfcTelecomAddress	Actor	1
	Approval	Approval	
	IfcApproval	Approval	0
	IfcApprovalActorRelationship	Approval	0
	IfcApprovalPropertyRelationship	Approval	0
	IfcApprovalRelationship	Approval	0
	Constraint	Constraint	
	(ABS) IfcConstraint	Constraint	n
	IfcConstraintAggregationRelationship	Constraint	0
	IfcConstraintClassificationRelationship	Constraint	0
	IfcConstraintRelationship	Constraint	0
	IfcMetric	Constraint	0
	IfcObjective	Constraint	0
	IfcPropertyConstraintRelationship	Constraint	0
	Cost	Cost	
	(ABS) IfcAppliedValue	Cost	n
	IfcAppliedValueRelationship	Cost	0
	IfcCostValue	Cost	0
	IfcCurrencyRelationship	Cost	0
	IfcEnvironmentalImpactValue	Cost	0
	IfcReferencesValueDocument	Cost	0
	DateTime	DateTime	
	IfcCalendarDate	DateTime	0
	IfcCoordinatedUniversalTimeOffset	DateTime	0
	IfcDateAndTime	DateTime	0
	IfcLocalTime	DateTime	0
	ExternalReference	ExternalReference	
	IfcClassification	ExternalReference	1
	IfcClassificationItem	ExternalReference	0
	IfcClassificationItemRelationship	ExternalReference	0

Architecture	BuildingServices	Structural
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	IfcSectionProperties	ProfileProperty	0
	IfcSectionReinforcementProperties	ProfileProperty	0
	IfcStructuralProfileProperties	ProfileProperty	0
	IfcStructuralSteelProfileProperties	ProfileProperty	0
	Profile	Profile	
	IfcArbitraryClosedProfileDef	Profile	1
	IfcArbitraryOpenProfileDef	Profile	1
	IfcArbitraryProfileDefWithVoids	Profile	1
	IfcAsymmetricIShapeProfileDef	Profile	1
	IfcCShapeProfileDef	Profile	1
	IfcCenterLineProfileDef	Profile	1
	IfcCircleHollowProfileDef	Profile	1
	IfcCircleProfileDef	Profile	1
	IfcCompositeProfileDef	Profile	1
	IfcCraneRailAShapeProfileDef	Profile	0
	IfcCraneRailFShapeProfileDef	Profile	0
	IfcDerivedProfileDef	Profile	1
	IfcEllipseProfileDef	Profile	1
	IfcIShapeProfileDef	Profile	1
	IfcLShapeProfileDef	Profile	1
	(ABS) IfcParameterizedProfileDef	Profile	y
	(ABS) IfcProfileDef	Profile	y
	IfcRectangleHollowProfileDef	Profile	1
	IfcRectangleProfileDef	Profile	1
	IfcRoundedRectangleProfileDef	Profile	1
	IfcTShapeProfileDef	Profile	1
	IfcTrapeziumProfileDef	Profile	0
	IfcUShapeProfileDef	Profile	1
	IfcZShapeProfileDef	Profile	1
	Property	Property	
	IfcComplexProperty	Property	1
	(ABS) IfcProperty	Property	y
	IfcPropertyBoundedValue	Property	1
	IfcPropertyDependencyRelationship	Property	0
	IfcPropertyEnumeratedValue	Property	1
	IfcPropertyEnumeration	Property	1
	IfcPropertyListValue	Property	1
	IfcPropertyReferenceValue	Property	0
	IfcPropertySingleValue	Property	1
	IfcPropertyTableValue	Property	1
	(ABS) IfcSimpleProperty	Property	y
	Quantity	Quantity	
	IfcPhysicalComplexQuantity	Quantity	0
	(ABS) IfcPhysicalQuantity	Quantity	n
	(ABS) IfcPhysicalSimpleQuantity	Quantity	n
	IfcQuantityArea	Quantity	0
	IfcQuantityCount	Quantity	0
	IfcQuantityLength	Quantity	0
	IfcQuantityTime	Quantity	0
	IfcQuantityVolume	Quantity	0
	IfcQuantityWeight	Quantity	0
	Representation	Representation	
	IfcGeometricRepresentationContext	Representation	1
	IfcGeometricRepresentationSubContext	Representation	1

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Utility	Utility	
IfcApplication	Utility	1
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Kernel	Kernel	
IfcActor	Kernel	0
(ABS) IfcControl	Kernel	n
IfcGroup	Kernel	1
(ABS) IfcObject	Kernel	y
(ABS) IfcObjectDefinition	Kernel	y
(ABS) IfcProcess	Kernel	n
(ABS) IfcProduct	Kernel	y
IfcProject	Kernel	1
(ABS) IfcPropertyDefinition	Kernel	y
IfcPropertySet	Kernel	1
(ABS) IfcPropertySetDefinition	Kernel	y
IfcProxy	Kernel	0
IfcRelAggregates	Kernel	1
(ABS) IfcRelAssigns	Kernel	y
IfcRelAssignsToActor	Kernel	0
IfcRelAssignsToControl	Kernel	0
IfcRelAssignsToGroup	Kernel	1
IfcRelAssignsToProcess	Kernel	0
IfcRelAssignsToProduct	Kernel	0
IfcRelAssignsToResource	Kernel	0
IfcRelAssociates	Kernel	1
IfcRelAssociatesClassification	Kernel	1
IfcRelAssociatesDocument	Kernel	0
IfcRelAssociatesLibrary	Kernel	0
(ABS) IfcRelConnects	Kernel	y
(ABS) IfcRelDecomposes	Kernel	y
(ABS) IfcRelDefines	Kernel	y
IfcRelDefinesByProperties	Kernel	1
IfcRelDefinesByType	Kernel	1
IfcRelNests	Kernel	0
IfcRelOverridesProperties	Kernel	0
IfcRelSequence	Kernel	0
(ABS) IfcRelationship	Kernel	y
(ABS) IfcResource	Kernel	n
(ABS) IfcRoot	Kernel	y
IfcTypeObject	Kernel	1
IfcTypeProduct	Kernel	1
Control	Control	
IfcPerformanceHistory	Control	0
IfcRelAssociatesApproval	Control	0
IfcRelAssociatesConstraint	Control	0
IfcTimeSeriesSchedule	Control	0
Process	Process	
IfcProcedure	Process	0
IfcRelAssignsTasks	Process	0
IfcScheduleTimeControl	Process	0
IfcTask	Process	0
(ABS) IfcWorkControl	Process	n
IfcWorkPlan	Process	0

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	IfcCurtainWall	SharedBld	1
	IfcCurtainWallType	SharedBld	1
	IfcDoor	SharedBld	1
	IfcDoorLiningProperties	SharedBld	1
	IfcDoorPanelProperties	SharedBld	1
	IfcDoorStyle	SharedBld	1
	IfcMember	SharedBld	1
	IfcMemberType	SharedBld	1
	IfcPlate	SharedBld	1
	IfcPlateType	SharedBld	1
	IfcRailing	SharedBld	1
	IfcRailingType	SharedBld	1
	IfcRamp	SharedBld	1
	IfcRampFlight	SharedBld	1
	IfcRampFlightType	SharedBld	1
	IfcRelConnectsPathElements	SharedBld	1
	IfcRoof	SharedBld	1
	IfcSlab	SharedBld	1
	IfcSlabType	SharedBld	1
	IfcStair	SharedBld	1
	IfcStairFlight	SharedBld	1
	IfcStairFlightType	SharedBld	1
	IfcWall	SharedBld	1
	IfcWallStandardCase	SharedBld	1
	IfcWallType	SharedBld	1
	IfcWindow	SharedBld	1
	IfcWindowLiningProperties	SharedBld	1
	IfcWindowPanelProperties	SharedBld	1
	IfcWindowStyle	SharedBld	1
	SharedBldgService	SharedBldgServic	
	IfcDistributionChamberElement	SharedBldgService	1
	IfcDistributionChamberElementType	SharedBldgService	1
	IfcDistributionControlElement	SharedBldgService	1
	(ABS) IfcDistributionControlElementType	SharedBldgService	y
	IfcDistributionFlowElement	SharedBldgService	1
	(ABS) IfcDistributionFlowElementType	SharedBldgService	y
	IfcDistributionPort	SharedBldgService	1
	IfcElectricalBaseProperties	SharedBldgService	0
	IfcEnergyConversionDevice	SharedBldgService	1
	(ABS) IfcEnergyConversionDeviceType	SharedBldgService	y
	IfcEnergyProperties	SharedBldgService	0
	IfcFlowController	SharedBldgService	1
	(ABS) IfcFlowControllerType	SharedBldgService	y
	IfcFlowFitting	SharedBldgService	1
	(ABS) IfcFlowFittingType	SharedBldgService	y
	IfcFlowMovingDevice	SharedBldgService	1
	(ABS) IfcFlowMovingDeviceType	SharedBldgService	y
	IfcFlowSegment	SharedBldgService	1
	(ABS) IfcFlowSegmentType	SharedBldgService	y
	IfcFlowStorageDevice	SharedBldgService	1
	(ABS) IfcFlowStorageDeviceType	SharedBldgService	y
	IfcFlowTerminal	SharedBldgService	1
	(ABS) IfcFlowTerminalType	SharedBldgService	y
	IfcFlowTreatmentDevice	SharedBldgService	1
	(ABS) IfcFlowTreatmentDeviceType	SharedBldgService	y
	IfcFluidFlowProperties	SharedBldgService	0
	IfcRelFlowControlElements	SharedBldgService	1
	IfcSoundProperties	SharedBldgService	0

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	IfcSpaceThermalLoadProperties	SharedBldgServic	0
	SharedComponents	SharedComponents	
	IfcChamferEdgeFeature	SharedComponents	0
	IfcDiscreteAccessory	SharedComponents	1
	IfcDiscreteAccessoryType	SharedComponents	1
	(ABS) IfcEdgeFeature	SharedComponents	n
	(ABS) IfcElementComponent	SharedComponents	y
	(ABS) IfcElementComponentType	SharedComponents	y
	IfcFastener	SharedComponents	1
	IfcFastenerType	SharedComponents	1
	IfcMechanicalFastener	SharedComponents	1
	IfcMechanicalFastenerType	SharedComponents	1
	IfcRoundedEdgeFeature	SharedComponents	0
	SharedFacilities	SharedFacilities	
	IfcAsset	SharedFacilities	0
	IfcFurnitureType	SharedFacilities	0
	IfcInventory	SharedFacilities	0
	IfcOccupant	SharedFacilities	0
	IfcRelOccupiesSpaces	SharedFacilities	0
	IfcServiceLife	SharedFacilities	0
	IfcServiceLifeFactor	SharedFacilities	0
	IfcSystemFurnitureElementType	SharedFacilities	0
	SharedMgm	SharedMgm	
	IfcCostItem	SharedMgm	0
	IfcCostSchedule	SharedMgm	0
	IfcProjectOrder	SharedMgm	0
	IfcProjectOrderRecord	SharedMgm	0
	IfcRelAssignsToProjectOrder	SharedMgm	0
	IfcRelAssociatesAppliedValue	SharedMgm	0
	IfcRelSchedulesCostItems	SharedMgm	0
	Architecture	Architecture	
	IfcPermeableCoveringProperties	Architecture	0
	IfcRelInteractionRequirements	Architecture	0
	IfcSpaceProgram	Architecture	0
	BuildingControls	BuildingControls	
	IfcActuatorType	BuildingControls	1
	IfcAlarmType	BuildingControls	1
	IfcControllerType	BuildingControls	1
	IfcFlowInstrumentType	BuildingControls	1
	IfcSensorType	BuildingControls	1
	ConstructionManagement	ConstructionManagement	
	IfcConstructionEquipmentResource	ConstructionManagement	0
	IfcConstructionMaterialResource	ConstructionManagement	0
	IfcConstructionProductResource	ConstructionManagement	0
	(ABS) IfcConstructionResource	ConstructionManagement	n
	IfcCrewResource	ConstructionManagement	0
	IfcLaborResource	ConstructionManagement	0
	IfcSubContractResource	ConstructionManagement	0
	Electrical	Electrical	
	IfcCableCarrierFittingType	Electrical	1
	IfcCableCarrierSegmentType	Electrical	1

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Actor	Actor
IfcActorRole	Actor
(ABS) IfcAddress	Actor
IfcOrganization	Actor
IfcPerson	Actor
IfcPersonAndOrganization	Actor
IfcPostalAddress	Actor
IfcTelecomAddress	Actor

ExternalReference	ExternalReference
IfcClassification	ExternalReference
IfcClassificationReference	ExternalReference
(ABS) IfcExternalReference	ExternalReference

GeometricConstraint	GeometricConstraint
IfcConnectionCurveGeometry	GeometricConstraint
(ABS) IfcConnectionGeometry	GeometricConstraint
IfcGridAxis	GeometricConstraint
IfcLocalPlacement	GeometricConstraint
(ABS) IfcObjectPlacement	GeometricConstraint

GeometricModel	GeometricModel
IfcBooleanClippingResult	GeometricModel
IfcBooleanResult	GeometricModel
IfcBoundingBox	GeometricModel
IfcExtrudedAreaSolid	GeometricModel
IfcFaceBasedSurfaceModel	GeometricModel
IfcFacetedBrep	GeometricModel
IfcGeometricCurveSet	GeometricModel
IfcGeometricSet	GeometricModel
IfcHalfSpaceSolid	GeometricModel
(ABS) IfcManifoldSolidBrep	GeometricModel
IfcPolygonalBoundedHalfSpace	GeometricModel
IfcRevolvedAreaSolid	GeometricModel
IfcShellBasedSurfaceModel	GeometricModel
(ABS) IfcSolidModel	GeometricModel
IfcSurfaceCurveSweptAreaSolid	GeometricModel
(ABS) IfcSweptAreaSolid	GeometricModel
IfcSweptDiskSolid	GeometricModel

Geometry	Geometry
IfcAxis1Placement	Geometry
IfcAxis2Placement2D	Geometry
IfcAxis2Placement3D	Geometry
(ABS) IfcBoundedCurve	Geometry
IfcCartesianPoint	Geometry
(ABS) IfcCartesianTransformationOperator	Geometry
IfcCartesianTransformationOperator2D	Geometry
IfcCartesianTransformationOperator2DnonUniform	Geometry
IfcCartesianTransformationOperator3D	Geometry
IfcCartesianTransformationOperator3DnonUniform	Geometry
IfcCircle	Geometry
IfcCompositeCurve	Geometry
IfcCompositeCurveSegment	Geometry
(ABS) IfcConic	Geometry
(ABS) IfcCurve	Geometry
IfcDirection	Geometry
(ABS) IfcElementarySurface	Geometry
IfcEllipse	Geometry
(ABS) IfcGeometricRepresentationItem	Geometry
IfcLine	Geometry
IfcMappedItem	Geometry
(ABS) IfcPlacement	Geometry
IfcPlane	Geometry
(ABS) IfcPoint	Geometry
IfcPolyline	Geometry
(ABS) IfcRepresentationItem	Geometry
IfcRepresentationMap	Geometry
(ABS) IfcSurface	Geometry
IfcTrimmedCurve	Geometry
IfcVector	Geometry

Material	Material
IfcMaterial	Material
IfcMaterialClassificationRelationship	Material
IfcMaterialLayer	Material
IfcMaterialLayerSet	Material
IfcMaterialLayerSetUsage	Material
IfcMaterialList	Material

Measure	Measure
IfcConversionBasedUnit	Measure
IfcDerivedUnit	Measure
IfcDerivedUnitElement	Measure
IfcDimensionalExponents	Measure
IfcMeasureWithUnit	Measure
(ABS) IfcNamedUnit	Measure
IfcSIUnit	Measure
IfcUnitAssignment	Measure

PresentationAppearance	PresentationAppearance
IfcCurveStyle	PresentationAppearance
(ABS) IfcPresentationStyle	PresentationAppearance
IfcPresentationStyleAssignment	PresentationAppearance
IfcStyledItem	PresentationAppearance
IfcSurfaceStyle	PresentationAppearance
IfcSurfaceStyleRendering	PresentationAppearance
IfcSurfaceStyleShading	PresentationAppearance

PresentationOrganization	PresentationOrganization
IfcPresentationLayerAssignment	PresentationOrganization
IfcPresentationLayerWithStyle	PresentationOrganization

Presentation	Presentation
IfcColourRgb	Presentation
(ABS) IfcColourSpecification	Presentation

Profile	Profile
IfcArbitraryClosedProfileDef	Profile
IfcArbitraryOpenProfileDef	Profile
IfcArbitraryProfileDefWithVoids	Profile
IfcAsymmetricIShapeProfileDef	Profile
IfcCShapeProfileDef	Profile
IfcCenterLineProfileDef	Profile
IfcCircleHollowProfileDef	Profile
IfcCircleProfileDef	Profile
IfcCompositeProfileDef	Profile
IfcDerivedProfileDef	Profile
IfcEllipseProfileDef	Profile
IfcIShapeProfileDef	Profile
IfcLShapeProfileDef	Profile
(ABS) IfcParameterizedProfileDef	Profile
(ABS) IfcProfileDef	Profile
IfcRectangleHollowProfileDef	Profile
IfcRectangleProfileDef	Profile
IfcRoundedRectangleProfileDef	Profile
IfcTShapeProfileDef	Profile
IfcUShapeProfileDef	Profile
IfcZShapeProfileDef	Profile

Property	Property
IfcComplexProperty	Property
(ABS) IfcProperty	Property
IfcPropertyBoundedValue	Property
IfcPropertyEnumeratedValue	Property
IfcPropertyEnumeration	Property
IfcPropertyListValue	Property
IfcPropertySingleValue	Property
IfcPropertyTableValue	Property
(ABS) IfcSimpleProperty	Property

Representation	Representation
IfcGeometricRepresentationContext	Representation
IfcGeometricRepresentationSubContext	Representation
IfcMaterialDefinitionRepresentation	Representation
IfcProductDefinitionShape	Representation
IfcProductRepresentation	Representation
IfcRepresentation	Representation
IfcRepresentationContext	Representation
(ABS) IfcShapeModel	Representation
IfcShapeRepresentation	Representation
(ABS) IfcStyleModel	Representation
IfcStyledRepresentation	Representation

Topology	Topology
IfcClosedShell	Topology
IfcConnectedFaceSet	Topology
IfcFace	Topology
IfcFaceBound	Topology
IfcFaceOuterBound	Topology
IfcLoop	Topology
IfcOpenShell	Topology
IfcPolyLoop	Topology
(ABS) IfcTopologicalRepresentationItem	Topology

Utility	Utility
IfcApplication	Utility
IfcOwnerHistory	Utility

Kernel	Kernel
IfcGroup	Kernel
(ABS) IfcObject	Kernel
(ABS) IfcObjectDefinition	Kernel
(ABS) IfcProduct	Kernel
IfcProject	Kernel
(ABS) IfcPropertyDefinition	Kernel
IfcPropertySet	Kernel
(ABS) IfcPropertySetDefinition	Kernel

IfcRelAggregates	Kernel
(ABS) IfcRelAssigns	Kernel
IfcRelAssignsToGroup	Kernel
IfcRelAssociates	Kernel
IfcRelAssociatesClassification	Kernel
(ABS) IfcRelConnects	Kernel
(ABS) IfcRelDecomposes	Kernel
(ABS) IfcRelDefines	Kernel
IfcRelDefinesByProperties	Kernel
IfcRelDefinesByType	Kernel
(ABS) IfcRelationship	Kernel
(ABS) IfcRoot	Kernel
IfcTypeObject	Kernel
IfcTypeProduct	Kernel

Product	Product
IfcBuilding	Product
(ABS) IfcBuildingElement	Product
IfcBuildingElementProxy	Product
IfcBuildingElementProxyType	Product
(ABS) IfcBuildingElementType	Product
IfcBuildingStorey	Product
IfcCovering	Product
IfcCoveringType	Product
IfcDistributionElement	Product
IfcDistributionElementType	Product
(ABS) IfcElement	Product
IfcElementAssembly	Product
(ABS) IfcElementType	Product
(ABS) IfcFeatureElement	Product
(ABS) IfcFeatureElementSubtraction	Product
IfcFurnishingElement	Product
IfcFurnishingElementType	Product
IfcGrid	Product
IfcOpeningElement	Product
(ABS) IfcPort	Product
IfcRelAssociatesMaterial	Product
IfcRelConnectsElements	Product
IfcRelConnectsPortToElement	Product
IfcRelConnectsPorts	Product
IfcRelConnectsWithRealizingElements	Product
IfcRelContainedInSpatialStructure	Product
IfcRelCoversBldgElements	Product
IfcRelFillsElement	Product
IfcRelServicesBuildings	Product
IfcRelVoidsElement	Product
IfcSite	Product
IfcSpace	Product
IfcSpaceType	Product
(ABS) IfcSpatialStructureElement	Product
(ABS) IfcSpatialStructureElementType	Product
IfcSystem	Product
IfcTransportElement	Product
IfcTransportElementType	Product
IfcZone	Product

SharedBld	SharedBld
IfcBeam	SharedBld
IfcBeamType	SharedBld
IfcColumn	SharedBld
IfcColumnType	SharedBld
IfcCurtainWall	SharedBld
IfcCurtainWallType	SharedBld
IfcDoor	SharedBld
IfcDoorLiningProperties	SharedBld
IfcDoorPanelProperties	SharedBld
IfcDoorStyle	SharedBld
IfcMember	SharedBld
IfcMemberType	SharedBld
IfcPlate	SharedBld
IfcPlateType	SharedBld
IfcRailing	SharedBld
IfcRailingType	SharedBld
IfcRamp	SharedBld
IfcRampFlight	SharedBld
IfcRampFlightType	SharedBld
IfcRelConnectsPathElements	SharedBld
IfcRoof	SharedBld
IfcSlab	SharedBld
IfcSlabType	SharedBld
IfcStair	SharedBld
IfcStairFlight	SharedBld
IfcStairFlightType	SharedBld
IfcWall	SharedBld
IfcWallStandardCase	SharedBld
IfcWallType	SharedBld
IfcWindow	SharedBld
IfcWindowLiningProperties	SharedBld
IfcWindowPanelProperties	SharedBld

IfcWindowStyle	SharedBld
SharedBldgService	SharedBldgServic
IfcDistributionChamberElement	SharedBldgService
IfcDistributionChamberElementType	SharedBldgService
IfcDistributionControlElement	SharedBldgService
(ABS) IfcDistributionControlElementType	SharedBldgService
IfcDistributionFlowElement	SharedBldgService
(ABS) IfcDistributionFlowElementType	SharedBldgService
IfcDistributionPort	SharedBldgService
IfcEnergyConversionDevice	SharedBldgService
(ABS) IfcEnergyConversionDeviceType	SharedBldgService
IfcFlowController	SharedBldgService
(ABS) IfcFlowControllerType	SharedBldgService
IfcFlowFitting	SharedBldgService
(ABS) IfcFlowFittingType	SharedBldgService
IfcFlowMovingDevice	SharedBldgService
(ABS) IfcFlowMovingDeviceType	SharedBldgService
IfcFlowSegment	SharedBldgService
(ABS) IfcFlowSegmentType	SharedBldgService
IfcFlowStorageDevice	SharedBldgService
(ABS) IfcFlowStorageDeviceType	SharedBldgService
IfcFlowTerminal	SharedBldgService
(ABS) IfcFlowTerminalType	SharedBldgService
IfcFlowTreatmentDevice	SharedBldgService
(ABS) IfcFlowTreatmentDeviceType	SharedBldgService
IfcRelFlowControlElements	SharedBldgServic
SharedComponents	SharedComponents
IfcDiscreteAccessory	SharedComponents
IfcDiscreteAccessoryType	SharedComponents
(ABS) IfcElementComponent	SharedComponents
(ABS) IfcElementComponentType	SharedComponents
IfcFastener	SharedComponents
IfcFastenerType	SharedComponents
IfcMechanicalFastener	SharedComponents
IfcMechanicalFastenerType	SharedComponents
BuildingControls	BuildingControls
IfcActuatorType	BuildingControls
IfcAlarmType	BuildingControls
IfcControllerType	BuildingControls
IfcFlowInstrumentType	BuildingControls
IfcSensorType	BuildingControls
Electrical	Electrical
IfcCableCarrierFittingType	Electrical
IfcCableCarrierSegmentType	Electrical
IfcCableSegmentType	Electrical
IfcElectricalCircuit	Electrical
IfcElectricApplianceType	Electrical
IfcElectricFlowStorageDeviceType	Electrical
IfcElectricGeneratorType	Electrical
IfcElectricMotorType	Electrical
IfcElectricTimeControlType	Electrical
IfcJunctionBoxType	Electrical
IfcLampType	Electrical
IfcLightFixtureType	Electrical
IfcMotorConnectionType	Electrical
IfcOutletType	Electrical
IfcProtectiveDeviceType	Electrical
IfcSwitchingDeviceType	Electrical
IfcTransformerType	Electrical
Hvac	Hvac
IfcAirTerminalBoxType	Hvac
IfcAirTerminalType	Hvac
IfcAirToAirHeatRecoveryType	Hvac
IfcBoilerType	Hvac
IfcChillerType	Hvac
IfcCoilType	Hvac
IfcCompressorType	Hvac
IfcCondenserType	Hvac
IfcCooledBeamType	Hvac
IfcCoolingTowerType	Hvac
IfcDamperType	Hvac
IfcDuctFittingType	Hvac
IfcDuctSegmentType	Hvac
IfcDuctSilencerType	Hvac
IfcEvaporativeCoolerType	Hvac
IfcEvaporatorType	Hvac
IfcFanType	Hvac
IfcFilterType	Hvac
IfcFlowMeterType	Hvac
IfcHeatExchangerType	Hvac
IfcHumidifierType	Hvac
IfcPipeFittingType	Hvac
IfcPipeSegmentType	Hvac
IfcPumpType	Hvac

IfcSpaceHeaterType	Hvac
IfcTankType	Hvac
IfcTubeBundleType	Hvac
IfcUnitaryEquipmentType	Hvac
IfcValveType	Hvac

PlumbingFireProtection	PlumbingFireProtection
IfcFireSuppressionTerminalType	PlumbingFireProtection
IfcSanitaryTerminalType	PlumbingFireProtection
IfcStackTerminalType	PlumbingFireProtection
IfcWasteTerminalType	PlumbingFireProtection

StructuralElements	StructuralElements
(ABS) IfcBuildingElementComponent	StructuralElements
IfcBuildingElementPart	StructuralElements
IfcFooting	StructuralElements
IfcPile	StructuralElements
IfcReinforcingBar	StructuralElements
(ABS) IfcReinforcingElement	StructuralElements
IfcReinforcingMesh	StructuralElements
IfcTendon	StructuralElements
IfcTendonAnchor	StructuralElements

Actor	Actor
IfcOrganizationRelationship	Actor
Approval	Approval
IfcApproval	Approval
IfcApprovalActorRelationship	Approval
IfcApprovalPropertyRelationship	Approval
IfcApprovalRelationship	Approval
Constraint	Constraint
(ABS) IfcConstraint	Constraint
IfcConstraintAggregationRelationship	Constraint
IfcConstraintClassificationRelationship	Constraint
IfcConstraintRelationship	Constraint
IfcMetric	Constraint
IfcObjective	Constraint
IfcPropertyConstraintRelationship	Constraint
Cost	Cost
(ABS) IfcAppliedValue	Cost
IfcAppliedValueRelationship	Cost
IfcCostValue	Cost
IfcCurrencyRelationship	Cost
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IfcReferencesValueDocument	Cost
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IfcClassificationItemRelationship	ExternalReference
IfcClassificationNotation	ExternalReference
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IfcDocumentInformation	ExternalReference
IfcDocumentInformationRelationship	ExternalReference
IfcDocumentReference	ExternalReference
IfcLibraryInformation	ExternalReference
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GeometricConstraint	GeometricConstraint
IfcConnectionPointEccentricity	GeometricConstraint
IfcConnectionPointGeometry	GeometricConstraint
IfcConnectionPortGeometry	GeometricConstraint
IfcConnectionSurfaceGeometry	GeometricConstraint
IfcGridPlacement	GeometricConstraint
IfcVirtualGridIntersection	GeometricConstraint
GeometricModel	GeometricModel
IfcBlock	GeometricModel
IfcBoxedHalfSpace	GeometricModel
(ABS) IfcCsgPrimitive3D	GeometricModel
IfcCsgSolid	GeometricModel
IfcFacetedBrepWithVoids	GeometricModel
IfcRectangularPyramid	GeometricModel
IfcRightCircularCone	GeometricModel
IfcRightCircularCylinder	GeometricModel
IfcSectionedSpine	GeometricModel
IfcSphere	GeometricModel
IfcSurfaceCurveSweptAreaSolid	GeometricModel
Geometry	Geometry
Ifc2DCompositeCurve	Geometry
(ABS) IfcBSplineCurve	Geometry
IfcBezierCurve	Geometry
IfcBoundedSurface	Geometry
IfcCurveBoundedPlane	Geometry
IfcOffsetCurve2D	Geometry
IfcOffsetCurve3D	Geometry
IfcPointOnCurve	Geometry
IfcPointOnSurface	Geometry
IfcRationalBezierCurve	Geometry
IfcRectangularTrimmedSurface	Geometry
IfcSurfaceOfLinearExtrusion	Geometry
IfcSurfaceOfRevolution	Geometry
(ABS) IfcSweptSurface	Geometry
MaterialProperty	MaterialProperty
IfcExtendedMaterialProperties	MaterialProperty
IfcFuelProperties	MaterialProperty
IfcGeneralMaterialProperties	MaterialProperty
IfcHygroscopicMaterialProperties	MaterialProperty
(ABS) IfcMaterialProperties	MaterialProperty
IfcMechanicalConcreteMaterialProperties	MaterialProperty
IfcMechanicalMaterialProperties	MaterialProperty

IfcMechanicalSteelMaterialProperties	MaterialProperty
IfcOpticalMaterialProperties	MaterialProperty
IfcProductsOfCombustionProperties	MaterialProperty
IfcRelaxation	MaterialProperty
IfcThermalMaterialProperties	MaterialProperty
IfcWaterProperties	MaterialProperty

Measure	Measure
IfcContextDependentUnit	Measure
IfcMonetaryUnit	Measure

PresentationAppearance	PresentationAppearance
IfcBlobTexture	PresentationAppearance
IfcCurveStyleFont	PresentationAppearance
IfcCurveStyleFontAndScaling	PresentationAppearance
IfcCurveStyleFontPattern	PresentationAppearance
IfcDraughtingPreDefinedCurveFont	PresentationAppearance
IfcExternallyDefinedHatchStyle	PresentationAppearance
IfcExternallyDefinedSurfaceStyle	PresentationAppearance
IfcFillAreaStyle	PresentationAppearance
IfcFillAreaStyleHatching	PresentationAppearance
IfcFillAreaStyleTileSymbolWithStyle	PresentationAppearance
IfcFillAreaStyleTiles	PresentationAppearance
IfcImageTexture	PresentationAppearance
IfcOneDirectionRepeatFactor	PresentationAppearance
IfcPixelTexture	PresentationAppearance
(ABS) IfcPreDefinedCurveFont	PresentationAppearance
IfcSurfaceStyleLighting	PresentationAppearance
IfcSurfaceStyleRefraction	PresentationAppearance
IfcSurfaceStyleWithTextures	PresentationAppearance
(ABS) IfcSurfaceTexture	PresentationAppearance
IfcSymbolStyle	PresentationAppearance
IfcTextStyle	PresentationAppearance
IfcTextStyleForDefinedFont	PresentationAppearance
IfcTextStyleTextModel	PresentationAppearance
IfcTextStyleWithBoxCharacteristics	PresentationAppearance
IfcTwoDirectionRepeatFactor	PresentationAppearance

PresentationDefinition	PresentationDefinition
IfcAnnotationCurveOccurrence	PresentationDefinition
IfcAnnotationFillArea	PresentationDefinition
IfcAnnotationFillAreaOccurrence	PresentationDefinition
(ABS) IfcAnnotationOccurrence	PresentationDefinition
IfcAnnotationSurface	PresentationDefinition
IfcAnnotationSurfaceOccurrence	PresentationDefinition
IfcAnnotationSymbolOccurrence	PresentationDefinition
IfcAnnotationTextOccurrence	PresentationDefinition
IfcDefinedSymbol	PresentationDefinition
IfcExternallyDefinedSymbol	PresentationDefinition
(ABS) IfcPreDefinedSymbol	PresentationDefinition
IfcTextLiteral	PresentationDefinition
IfcTextLiteralWithExtent	PresentationDefinition
(ABS) IfcTextureCoordinate	PresentationDefinition
IfcTextureCoordinateGenerator	PresentationDefinition
IfcTextureMap	PresentationDefinition
IfcTextureVertex	PresentationDefinition
IfcVertexBasedTextureMap	PresentationDefinition

PresentationDimensioning	PresentationDimensioning
IfcAngularDimension	PresentationDimensioning
IfcDiameterDimension	PresentationDimensioning
IfcDimensionCalloutRelationship	PresentationDimensioning
IfcDimensionCurve	PresentationDimensioning
IfcDimensionCurveDirectedCallout	PresentationDimensioning
IfcDimensionCurveTerminator	PresentationDimensioning
IfcDimensionPair	PresentationDimensioning
IfcDraughtingCallout	PresentationDimensioning
IfcDraughtingCalloutRelationship	PresentationDimensioning
IfcLinearDimension	PresentationDimensioning
IfcPreDefinedDimensionSymbol	PresentationDimensioning
IfcPreDefinedPointMarkerSymbol	PresentationDimensioning
IfcPreDefinedTerminatorSymbol	PresentationDimensioning
IfcProjectionCurve	PresentationDimensioning
IfcRadiusDimension	PresentationDimensioning
IfcStructuredDimensionCallout	PresentationDimensioning
IfcTerminatorSymbol	PresentationDimensioning

PresentationOrganization	PresentationOrganization
IfcLightDistributionData	PresentationOrganization
IfcLightIntensityDistribution	PresentationOrganization
(ABS) IfcLightSource	PresentationOrganization
IfcLightSourceAmbient	PresentationOrganization
IfcLightSourceDirectional	PresentationOrganization
IfcLightSourceGoniometric	PresentationOrganization
IfcLightSourcePositional	PresentationOrganization
IfcLightSourceSpot	PresentationOrganization

Presentation	Presentation
IfcDraughtingPreDefinedColour	Presentation

IfcDraughtingPreDefinedTextFont	Presentation
IfcExternallyDefinedTextFont	Presentation
IfcPlanarBox	Presentation
IfcPlanarExtent	Presentation
(ABS) IfcPreDefinedColour	Presentation
(ABS) IfcPreDefinedItem	Presentation
(ABS) IfcPreDefinedTextFont	Presentation
IfcTextStyleFontModel	Presentation

ProfileProperty	ProfileProperty
IfcGeneralProfileProperties	ProfileProperty
(ABS) IfcProfileProperties	ProfileProperty
IfcReinforcementBarProperties	ProfileProperty
IfcRibPlateProfileProperties	ProfileProperty
IfcSectionProperties	ProfileProperty
IfcSectionReinforcementProperties	ProfileProperty
IfcStructuralProfileProperties	ProfileProperty
IfcStructuralSteelProfileProperties	ProfileProperty

Profile	Profile
IfcCraneRailAShapeProfileDef	Profile
IfcCraneRailFShapeProfileDef	Profile
IfcTrapeziumProfileDef	Profile

Property	Property
IfcPropertyDependencyRelationship	Property
IfcPropertyReferenceValue	Property

Quantity	Quantity
IfcPhysicalComplexQuantity	Quantity
(ABS) IfcPhysicalQuantity	Quantity
(ABS) IfcPhysicalSimpleQuantity	Quantity
IfcQuantityArea	Quantity
IfcQuantityCount	Quantity
IfcQuantityLength	Quantity
IfcQuantityTime	Quantity
IfcQuantityVolume	Quantity
IfcQuantityWeight	Quantity

Representation	Representation
IfcShapeAspect	Representation
IfcTopologyRepresentation	Representation

StructuralLoad	StructuralLoad
(ABS) IfcBoundaryCondition	StructuralLoad
IfcBoundaryEdgeCondition	StructuralLoad
IfcBoundaryFaceCondition	StructuralLoad
IfcBoundaryNodeCondition	StructuralLoad
IfcBoundaryNodeConditionWarping	StructuralLoad
IfcFailureConnectionCondition	StructuralLoad
IfcSlippageConnectionCondition	StructuralLoad
(ABS) IfcStructuralConnectionCondition	StructuralLoad
(ABS) IfcStructuralLoad	StructuralLoad
IfcStructuralLoadLinearForce	StructuralLoad
IfcStructuralLoadPlanarForce	StructuralLoad
IfcStructuralLoadSingleDisplacement	StructuralLoad
IfcStructuralLoadSingleDisplacementDistortion	StructuralLoad
IfcStructuralLoadSingleForce	StructuralLoad
IfcStructuralLoadSingleForceWarping	StructuralLoad
(ABS) IfcStructuralLoadStatic	StructuralLoad
IfcStructuralLoadTemperature	StructuralLoad

TimeSeries	TimeSeries
IfcIrregularTimeSeries	TimeSeries
IfcIrregularTimeSeriesValue	TimeSeries
IfcRegularTimeSeries	TimeSeries
(ABS) IfcTimeSeries	TimeSeries
IfcTimeSeriesReferenceRelationship	TimeSeries
IfcTimeSeriesValue	TimeSeries

Topology	Topology
IfcEdge	Topology
IfcEdgeCurve	Topology
IfcEdgeLoop	Topology
IfcFaceSurface	Topology
IfcOrientedEdge	Topology
IfcPath	Topology
IfcSubedge	Topology
IfcVertex	Topology
IfcVertexLoop	Topology
IfcVertexPoint	Topology

Utility	Utility
IfcTable	Utility
IfcTableRow	Utility

Kernel	Kernel
IfcActor	Kernel
(ABS) IfcControl	Kernel

(ABS) IfcProcess	Kernel
IfcProxy	Kernel
IfcRelAssignsToActor	Kernel
IfcRelAssignsToControl	Kernel
IfcRelAssignsToProcess	Kernel
IfcRelAssignsToProduct	Kernel
IfcRelAssignsToResource	Kernel
IfcRelAssociatesDocument	Kernel
IfcRelAssociatesLibrary	Kernel
IfcRelNests	Kernel
IfcRelOverridesProperties	Kernel
IfcRelSequence	Kernel
(ABS) IfcResource	Kernel

Control	Control
IfcPerformanceHistory	Control
IfcRelAssociatesApproval	Control
IfcRelAssociatesConstraint	Control
IfcTimeSeriesSchedule	Control

Process	Process
IfcProcedure	Process
IfcRelAssignsTasks	Process
IfcScheduleTimeControl	Process
IfcTask	Process
(ABS) IfcWorkControl	Process
IfcWorkPlan	Process
IfcWorkSchedule	Process

Product	Product
IfcAnnotation	Product
IfcElectricalElement	Product
IfcElementQuantity	Product
IfcEquipmentElement	Product
(ABS) IfcFeatureElementAddition	Product
IfcProjectionElement	Product
IfcRelCoversSpaces	Product
IfcRelProjectsElement	Product
IfcRelReferencedInSpatialStructure	Product
IfcRelSpaceBoundary	Product
IfcVirtualElement	Product

SharedBldgService	SharedBldgServic
IfcElectricalBaseProperties	SharedBldgService
IfcEnergyProperties	SharedBldgService
IfcFluidFlowProperties	SharedBldgService
IfcSoundProperties	SharedBldgService
IfcSoundValue	SharedBldgService
IfcSpaceThermalLoadProperties	SharedBldgService

SharedComponents	SharedComponents
IfcChamferEdgeFeature	SharedComponents
(ABS) IfcEdgeFeature	SharedComponents
IfcRoundedEdgeFeature	SharedComponents

SharedFacilities	SharedFacilities
IfcAsset	SharedFacilities
IfcFurnitureType	SharedFacilities
IfcInventory	SharedFacilities
IfcOccupant	SharedFacilities
IfcRelOccupiesSpaces	SharedFacilities
IfcServiceLife	SharedFacilities
IfcServiceLifeFactor	SharedFacilities
IfcSystemFurnitureElementType	SharedFacilities

SharedMgm	SharedMgm
IfcCostItem	SharedMgm
IfcCostSchedule	SharedMgm
IfcProjectOrder	SharedMgm
IfcProjectOrderRecord	SharedMgm
IfcRelAssignsToProjectOrder	SharedMgm
IfcRelAssociatesAppliedValue	SharedMgm
IfcRelSchedulesCostItems	SharedMgm

Architecture	Architecture
IfcPermeableCoveringProperties	Architecture
IfcRelInteractionRequirements	Architecture
IfcSpaceProgram	Architecture

ConstructionManagement	ConstructionManagement
IfcConstructionEquipmentResource	ConstructionManagement
IfcConstructionMaterialResource	ConstructionManagement
IfcConstructionProductResource	ConstructionManagement
(ABS) IfcConstructionResource	ConstructionManagement
IfcCrewResource	ConstructionManagement
IfcLaborResource	ConstructionManagement
IfcSubContractResource	ConstructionManagement

Electrical	Electrical
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IfcElectricalCircuit	Electrical
IfcElectricDistributionPoint	Electrical
IfcElectricHeaterType	Electrical
FacilitiesManagement	FacilitiesManagement
IfcActionRequest	FacilitiesManagement
IfcCondition	FacilitiesManagement
IfcConditionCriterion	FacilitiesManagement
IfcEquipmentStandard	FacilitiesManagement
IfcFurnitureStandard	FacilitiesManagement
IfcMove	FacilitiesManagement
IfcOrderAction	FacilitiesManagement
IfcPermit	FacilitiesManagement
Hvac	Hvac
IfcGasTerminalType	Hvac
IfcVibrationIsolatorType	Hvac
StructuralElements	StructuralElements
IfcReinforcementDefinitionProperties	StructuralElements
StructuralAnalysis	StructuralAnalysis
IfcRelAssociatesProfileProperties	StructuralAnalysis
IfcRelConnectsStructuralActivity	StructuralAnalysis
IfcRelConnectsStructuralElement	StructuralAnalysis
IfcRelConnectsStructuralMember	StructuralAnalysis
IfcRelConnectsWithEccentricity	StructuralAnalysis
(ABS) IfcStructuralAction	StructuralAnalysis
(ABS) IfcStructuralActivity	StructuralAnalysis
IfcStructuralAnalysisModel	StructuralAnalysis
(ABS) IfcStructuralConnection	StructuralAnalysis
IfcStructuralCurveConnection	StructuralAnalysis
IfcStructuralCurveMember	StructuralAnalysis
IfcStructuralCurveMemberVarying	StructuralAnalysis
(ABS) IfcStructuralItem	StructuralAnalysis
IfcStructuralLinearAction	StructuralAnalysis
IfcStructuralLinearActionVarying	StructuralAnalysis
IfcStructuralLoadGroup	StructuralAnalysis
(ABS) IfcStructuralMember	StructuralAnalysis
IfcStructuralPlanarAction	StructuralAnalysis
IfcStructuralPlanarActionVarying	StructuralAnalysis
IfcStructuralPointAction	StructuralAnalysis
IfcStructuralPointConnection	StructuralAnalysis
IfcStructuralPointReaction	StructuralAnalysis
(ABS) IfcStructuralReaction	StructuralAnalysis
IfcStructuralResultGroup	StructuralAnalysis
IfcStructuralSurfaceConnection	StructuralAnalysis
IfcStructuralSurfaceMember	StructuralAnalysis
IfcStructuralSurfaceMemberVarying	StructuralAnalysis